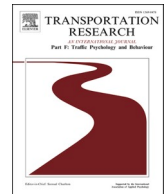




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Driver's avoidance characteristics to hazardous situations: A driving simulator study

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ABSTRACT

When analyzing the causes of an accident, it is critical to determine whether the driver could have prevented the accident. In previous studies on the reaction times of drivers, the definition and values of reaction times vary, so applying reaction time is difficult. In such analysis, the driver's reaction time from perception is required to determine whether the driver could have prevented the accident, but past studies are difficult to utilize in accident analysis as reaction time measurements were taken after the occurrence of hazardous situations. In this study, 93 subjects from age groups ranging from 20 s to 40 s participated in an experiment inside a full-scale driving simulator, to determine reaction time values that can be practically applied to accident analysis. A total of 4 hazardous accident situations were reproduced, including driving over the centerline, pedestrian jaywalking, a vehicle cutting in, and intersection traffic signal violation. The Time-To-Collision (TTC) was 2.5 s and the driving speed was set to the common city road speed limits of 60 and 80 km/h. An eye tracker was used to determine the driver's Saccade Latency (SL) during hazardous situations. Brake Reaction Time from Perception (BRT_p), Steer Reaction Time from Perception (SRT_p), and Driver Reaction Time from Perception (DRT_p) were derived, and the measurements were statistically analyzed to investigate differences by age group, gender, speed, and type of hazardous situation. Most participants were found to avoid collisions by braking first rather than steering for the presented hazardous situations, except for the cutting in situation. Also, to determine a reaction time that would cover most drivers, the 85th percentile of DRT_p was calculated. The 85th percentile of DRT_p was in the range of 0.550 – 0.800 s. Specifically for each hazardous situation, it was 0.650 s for driving over the centerline, 0.800 s for the pedestrian jaywalking, 0.660 s for cutting in, and 0.550 s for the intersection traffic signal violation. For all 4 hazardous situations combined, the 85th percentile of DRT_p was 0.646 s. The findings can be utilized to determine the driver's likelihood of avoiding accidents when faced with similar hazardous situations.

1. Introduction

According to the World Health Organization (2018), there are around 1.35 million deaths and 50 million injuries worldwide

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Nomenclature

SL	Saccade Latency, s
BRT _p	Brake Reaction Time after Perception, s
SRT _p	Steer Reaction Time after Perception, s
DRT _p	Driver Reaction Time after Perception, s

annually from traffic accidents. Statistics reveal that traffic accidents are the 8th highest cause of death for all age groups, while traffic accidents are the number 1 cause of death for young people between the ages of 5–29. In the case of South Korea, 3,781 people died and 323,037 people were injured from traffic accidents in 2018 (Korea Road Traffic Authority, 2019). That year, traffic accidents were the 11st highest cause of death for in South Korea, and consistent with the announcement from the World Health Organization that traffic accidents are a leading cause of death for young age groups, traffic accidents were the 2nd highest cause of death for young people between the ages of 1–9 and in their 20 s and the 3rd highest for young people in their 10 s (Statistics Korea, 2019).

Most traffic accidents are caused by human factors, given the fact that people directly drive cars. According to the study by Pakgohar et al. (2011), 97.5% of all traffic accident causes were related to human factors. In the event of such an accident, an accident is reconstructed to analyze how it happened. In particular, human reaction time by the driver is an important factor in determining whether the driver could have avoided the accident when it happened. Here, reaction time refers to the time consumed between the driver's perception of a hazardous situation while driving until the driver brakes or steers in an attempt to avoid collision.

Previous research has investigated reaction time. For example, Olson and Sivak (1986) carried out a crest vertical curve vehicle test with 15 elderly drivers and 49 non-elderly driver subjects to measure the time from discovery of an unexpected obstacle to operation of the brake pedal, and their results revealed that the 95th percentile of both groups was approximately 1.6 s and there was no significant difference in reaction time between the non-elderly and elderly drivers. Olson (1989) showed that in most data with a reasonably clear stimulus and a fairly straightforward situation, 85% of drivers reacted within 1.5 s after hazard onset. However, because there are numerous factors that affect reaction time, it is difficult to make blanket conclusions regarding reaction time and it was proposed that the reaction time has to be evaluated differently for different hazardous situations.

Uno and Hiramatsu (2001) reproduced a hazardous situation in a simulator with 20 elderly drivers aged 65 and above and 20 non-elderly drivers aged below 60 to compare their avoidance behaviors in a hazardous situation. The time at which a vehicle appears suddenly from the left side of the driver in an intersection was varied. The time it took for elderly drivers to avoid an accident after the occurrence of the hazardous situation was 1.35–2.7 s, which was longer than the time of 1.2–2.4 s observed for the non-elderly drivers. Moreover, as the time available for avoidance decreased, the reaction time of the drivers also decreased. Here, the maximum of the average reaction time for each appearance time was 0.8 s.

Among past studies related to drivers' age groups, Islam and Ozkul (2019) found that commercial truck drivers had different risk factors in large-truck fatal crashes when categorized by age into four groups: under 30, 30–49, 50–65, and above 65. Islam (2021) conducted a study on the effect of motorcyclists' age on injury severities in single-motorcycle crashes, and showed through a logit analysis that age group influenced motorcyclists' safety perception when motorcyclists were divided into three age groups: under 30, 30–49, and above 50.

Jurecki and Stańczyk (2014) carried out a vehicle experiment to investigate the reaction time of the driver in which a pedestrian suddenly appeared on the left and right sides of the driver with varying appearance times, so that the TTC varied from 0.6 s to 3.0 s. The results showed that the reaction time of the driver decreased with decreasing TTC or decreasing time for the driver to react. This finding was in agreement with the aforementioned research results of Uno and Hiramatsu (2001), and the average range of the brake reaction time was 0.65–1.6 s. Thus, Jurecki and Stańczyk (2014) proposed that the reaction time be applied differently depending on when the hazardous situation occurs, even if the situation is the same.

Toxopeus et al. (2018a) used a simulator to conduct a hazardous situation experiment with 26 participants where a bicycle appears on the right side of the driver at an intersection. The resulting average brake reaction time was 1.92 s, average steer reaction time was 1.95 s, and the response of 80% of the participants prioritized braking over steering. Toxopeus et al. (2018b) reproduced the hazardous situation using a simulator where a pedestrian comes running at mid-block on the left side of the driver and the reaction time was measured. It was found that the average brake reaction time was 1.43 s and the average steer reaction time was 1.78 s.

Attalla et al. (2018) recruited 98 participants in a simulator experiment reproducing the situation where a vehicle stops on the other side of an intersection and makes a left turn, and a vehicle that does not stop at the intersection and continues to make a left turn. The reaction time of the participants when facing the corresponding vehicle was measured. The results revealed that the average brake reaction time was 2.70 s when the vehicle made a left turn after stopping, and the average brake reaction time was 2.02 s for the situation where the vehicle made a left turn without stopping. Also, it was found that the reaction time was longer for female drivers than male drivers, but there was no significant difference in the accident rate.

Pawar and Patil (2017) tested for four events at unsignalized intersections using a driving simulator, and obtained measurements for response time before possible conflict (RTPC). The 10th and 90th percentile RTPC values varied from 6.3 to 9.4 s and 0.1 to 1.8 s, respectively. Akinapalli and Pawar (2021) performed a mock-up control study for motorized two-wheeler riders to examine brake reaction times (BRT) and evasive action maneuver. The 90th percentile BRT for the unexpected and surprise events were 3.62 and 1.6 s, respectively, thereby establishing regression models of BRT with approach speeds and time to collision (TTC).

Accidents may be caused not only by hazardous situations, but also drivers' circumstances. Choudhary et al. (2020) examined

distracted driving performance of 90 drivers using Structural Equation Modelling (SEM), and found that texting had the highest impact on crash risk. Khan et al. (2020) identified different weather conditions, sleep deprivation, unsafe lane changes, and night-time driving as other factors contributing to traffic accidents, and Ramiani and Shirazian (2020) showed that season, daytime light, age group, motorcycles, and pedestrians were among factors influencing road freight accidents.

Additionally, reaction time was used to evaluate the suitability of road design standards. The American Association of State Highway and Transportation Officials (AASHTO) (2018) reported that the total time for drivers to sufficiently perceive and react to a hazardous situation was 2.5 s, and this result has been employed in road design standards worldwide.

As investigated in the previous research, the occurrence time of a hazardous situation and the corresponding results can vary even for similar situations, so applying a specific value for traffic accident analysis is difficult. Most studies have also started to measure the perception reaction time starting from the timepoint at which the hazardous situation begins. This makes it difficult to determine the actual time when the hazardous situation was perceived by the driver, as well as how much time was consumed in making an avoidance response after perception, including steering or braking. However, the driver's time of perception of the hazardous situation is extremely important when analyzing actual traffic accidents. This is because different drivers become aware of hazardous situations at different times, which in turn affects their likelihood of avoiding accidents. This study used an eye tracker to obtain data for the actual accident analysis, and determined the time taken for drivers to perceive hazardous situations after their occurrence. Steer reaction and brake reaction times after perception were measured and analyzed to determine whether there were significant differences by age, gender, speed, and type of hazardous situation. To accurately assess reaction time during regular driving behavior, the drivers were not assigned any tasks other than driving.

Section 2, Methods, which covers the general experimental design, presents the independent variables thought to possibly have an impact on the accident avoidance of drivers, and describes the age and gender of participants. The four hazardous situations of driving over the centerline, pedestrian jaywalking, cutting in, and intersection traffic signal violation were introduced, and the dependent variables derived through the experiment, specifically, SL, BRT_p, SRT_p, and DRT_p, were described. Section 3, Results & Discussion, discusses the dependent variables by age group, gender, speed, and type of hazardous situation, and describes the 85th percentile values of BRT_p and DRT_p, which reveal how the majority of drivers react. Section 4, Conclusion, highlights useful values for accident analysis, and describes matters to be considered in future research.

2. Methods

2.1. Independent variables

The independent variables in this study, which are thought to possibly have an impact on the reaction time of the driver, were demographic characteristics, speed, and the hazardous situation, as shown in Table 1. The demographic characteristics of the driver were age group and gender. The age groups were categorized into 20 s, 30 s, and 40 s while gender was categorized into male and female. The speed was set to 60 km/h and 80 km/h, which are common city road speed limits. For the 4 hazardous situations of driving over the centerline, cutting in, pedestrian jaywalking, and intersection traffic signal violation that frequently lead to traffic accidents were considered.

2.2. Participants

The major independent variables related to the participants in this study were age group and gender, so as much as possible the participants were recruited considering uniform age group and gender distribution. The experiment was carried out with a total of 105 subjects as shown in Table 2 and data was acquired for 93 participants, after excluding 12 participants who quit due to motion sickness. Participants were required to be driver's license holders with actual driving experience of at least 1 year and not wear glasses for the data acquisition using an eye tracker. This study was approved by the Kookmin University Institutional Review Board (KMU-201914-HR-205) and the regulations of the board were adhered to when carrying out the experiment. Participants in their 20 s were recruited from within the university, while those in their 30 s and 40 s were recruited through an external agency. Upon completion, the

Table 1
Independent variables.

Main category	Sub category	Classification
Demographic Characteristics	Age group	20 s
		30 s
		40 s
	Gender	Male
Speed		Female
		60 km/h
		80 km/h
Hazardous situation		Driving over the centerline
		Pedestrian jaywalking
		Cutting in
		Intersection traffic signal violation

Table 2

Experiment participant number according to the age group and gender.

Age group	Gender	Experiment completion	Exclusion	Data acquisition
20 s	Male	15	0	15
	Female	17	1	16
30 s	Male	15	0	15
	Female	16	1	15
40 s	Male	28	6	22
	Female	14	4	10
Total		105	12	93

participants each received \$90 as compensation.

2.3. Hazardous situations

The hazardous situations for this experiment were composed of the 4 selected situations that frequently lead to traffic accidents. Based on the time of 2.5 s needed for the driver to perceive a hazardous situation and sufficiently respond, as presented by the AASHTO (2011), all of the hazardous situations were set so that each hazard occurred when TTC was 2.5 s.

2.3.1. Driving over the centerline

The driving over the centerline situation was reproduced so that it occurred on a two-lane road as shown in Fig. 1. It was set so that 2 vehicles drive one after the other on the lane opposite of the subject vehicle. The preceding vehicle is moving at 30 km/h and the object vehicle followed the preceding vehicle at a speed of 60 km/h. When the TTC of the subject and object vehicles reached 2.5 s, the object vehicle drives over the centerline in an attempt to drive past the preceding vehicle and faces the subject vehicle.

2.3.2. Pedestrian jaywalking

The pedestrian jaywalking situation was reproduced so that it occurred on a four-lane road as shown in Fig. 2 and there are multiple vehicles parked on the right-hand side of the road in the direction of the subject vehicle. When the TTC of the subject vehicle and pedestrian reaches 2.5 s, a child pedestrian appears from between the parked cars on the right at a speed of 5 km/h. This hazardous situation referenced the Test Protocol – AEB VRU Systems of Euro NCAP (2019).

2.3.3. Cutting in

The cutting in situation was reproduced so that it occurred on a four-lane road as shown in Fig. 3 and there are multiple parked cars on the right-hand side of the road of the subject vehicle. When the TTC of the subject and object vehicles reached 2.5 s, the object

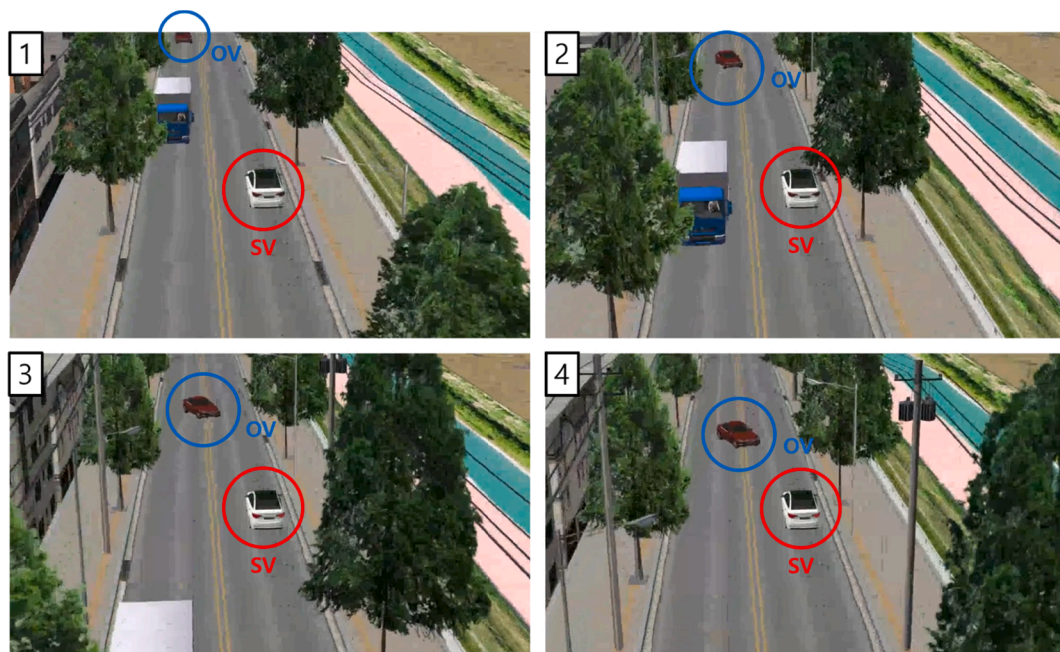


Fig. 1. Driving over the centerline reproduced in the simulator environment.

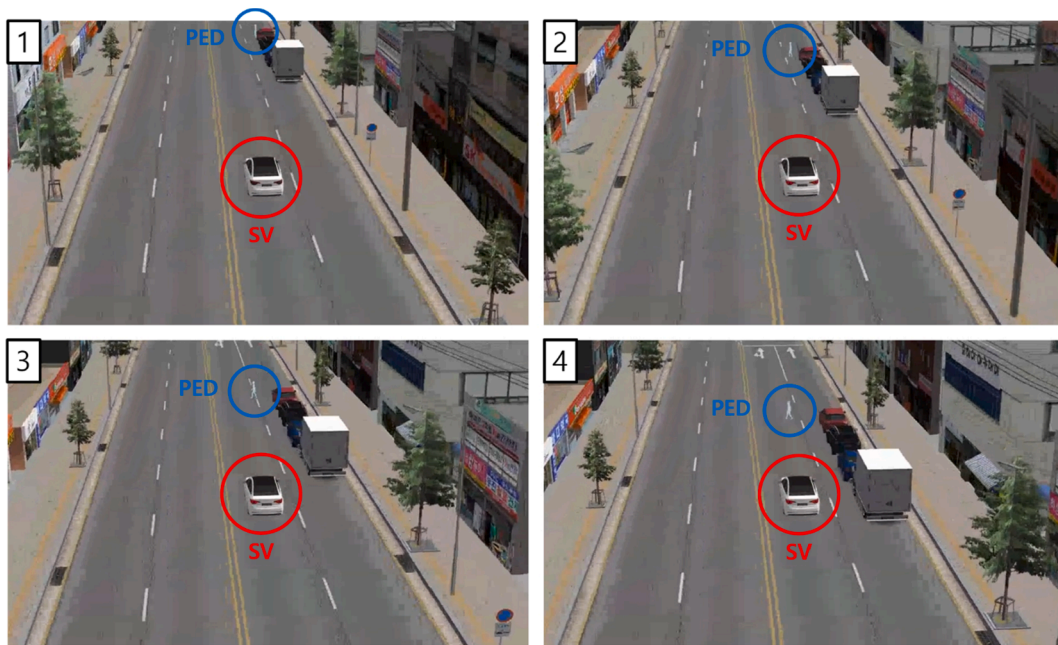


Fig. 2. Pedestrian jaywalking reproduced in the simulator environment.

vehicle that was at a stop on the right-hand side of the road suddenly accelerates with the left directional indicator on, to enter the lane of the subject vehicle.

2.3.4. Intersection traffic signal violation

The intersection traffic signal violation situation was reproduced so that it occurred at a crossroad intersection as shown in Fig. 4. The intersection traffic signal was set to turn green so that the subject vehicle can continue driving forward without being interrupted by the signal when passing through the intersection. The object vehicle is located at the outermost lane of the road on the right of the driving direction of the subject vehicle and a bus is at a stop on the innermost lane so that the experiment participant is unable to see the object vehicle. When the TTC of the subject and object vehicles reached 2.5 s, the object vehicle on the right of the intersection

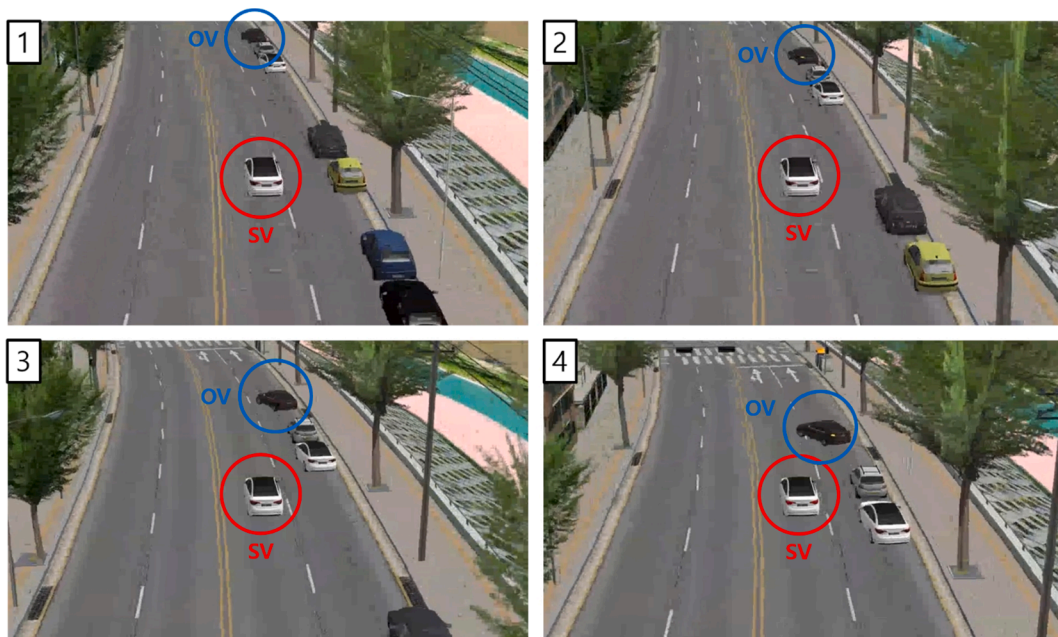


Fig. 3. Cutting in situation reproduced in the simulator environment.

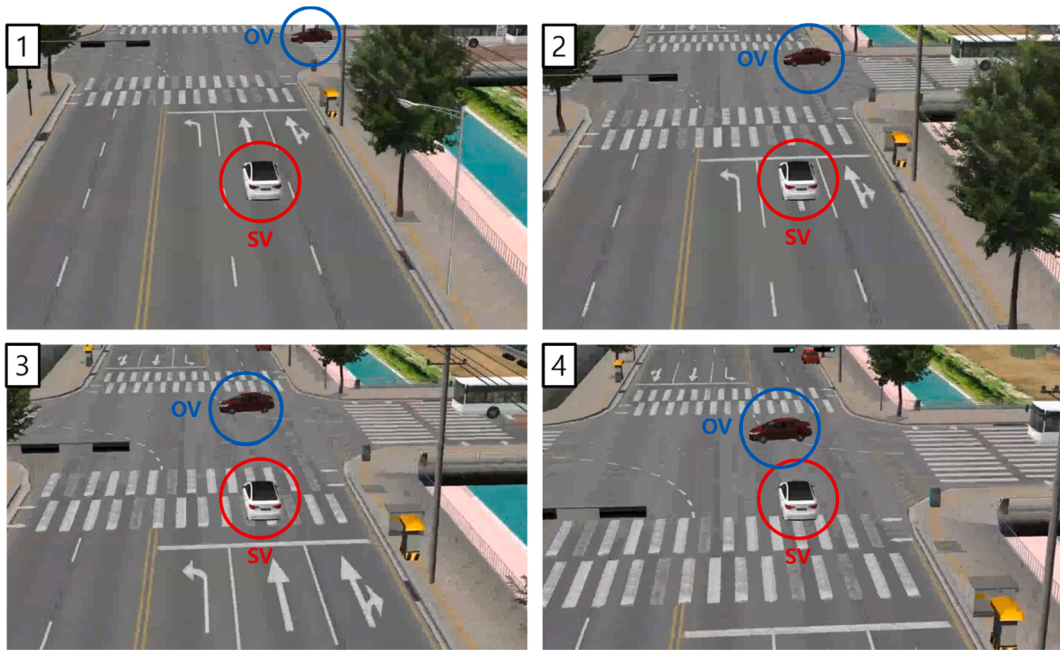


Fig. 4. Intersection traffic signal violation situation reproduced in the simulator environment.

violates the traffic signal at a speed of 60 km/h and makes a sudden stop in the center of the intersection.

2.4. Dependent variables

The dependent variables were set to be the driving time between the point of hazard onset to the point of hazard perception, and the time from hazard perception to operating the brake pedal and steering wheel in avoidance after cognition of the hazardous situation and making a decision. For the reaction times used in previous research, time was measured starting from hazard onset, and were designated as BRT, SRT, and DRT depending on the reaction method, including braking and steering (Markkula et al., 2016; D'Addario and Donmez, 2019; Attalla et al., 2018).

In this study, the times from hazardous situation perception by the experiment participant to the reaction afterwards were measured and designated BRT_p, SRT_p, and DRT_p to distinguish them from the existing dependent variable names. As an example, Fig. 5 shows a schematic diagram of the variables, and Fig. 6 shows the dependent variables in the situation of evading a vehicle cutting in while driving at 60 km/h.

Saccade Latency (SL)

After hazard onset during driving, when the experiment participant perceives the hazard, the gaze of the experimental participant moves (saccade) to where the hazardous situation is located. SL was defined as the time at the last fixation at hazard onset before moving to the hazardous situation object (D'Addario and Donmez, 2019; Recarte and Nunes, 2003). Fig. 6(a) shows the eye tracker data of an experiment participant where the gaze of the participant is fixed forward while driving, and when the gaze of the participant begins to move (saccade) to the object vehicle cutting in after the occurrence of the hazardous situation. The time consumed to this point is SL.

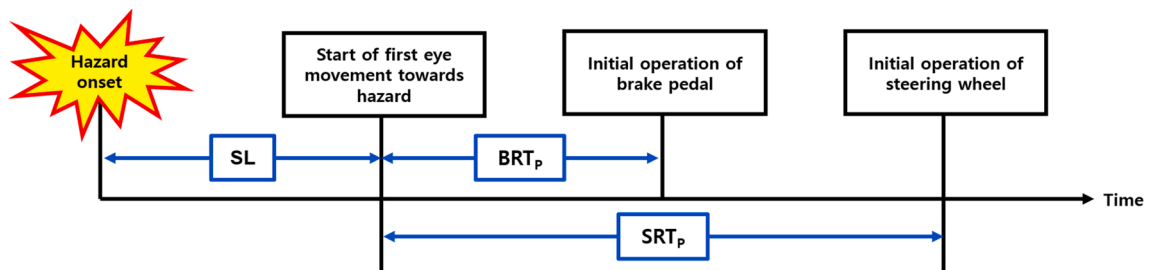


Fig. 5. Scheme of dependent variables. Adapted from D'Addario and Donmez (2019). The order of operation of the brakes and steering wheel may vary between individual, and in the Figure above, the DRT_p is BRT_p.

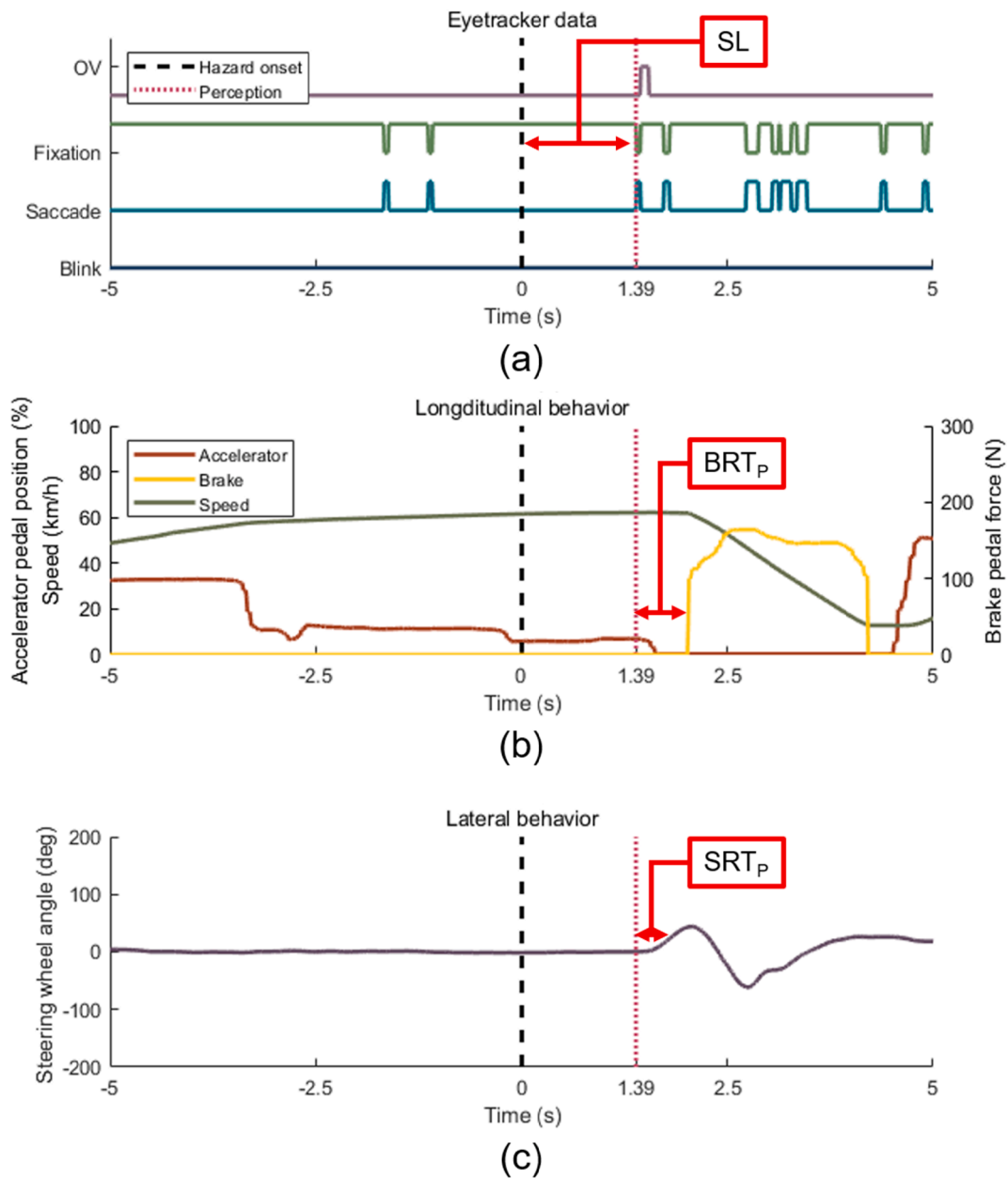


Fig. 6. Driver's behavior while driving 60 km/h in cutting in situation.

Brake Reaction Time from Perception (BRT_P)

The brake reaction time from perception is defined as the time from hazard onset perception by the experiment participant to the operation of the brake pedal for avoidance after cognition of the hazardous situation, followed by making a decision. The commonly used BRT is the time from hazard onset to the first operation of the brake pedal (SAE, 2015) but in this study, SL was excluded from this conventional BRT in order to determine the reaction time after perception. Fig. 6(b) shows the longitudinal behavior of the experiment participant where the accelerator pedal displacement, brake pedal effort force, and speed can be observed. The time taken for participants to step on the brake pedal after perception was denoted as BRT_P.

Steer Reaction Time from Perception (SRT_P)

The steer reaction time from perception is defined as the time after hazard onset perception by the experiment participant to the operation of the steering wheel for avoidance after cognition of the hazardous situation followed by making a decision. In order to observe the urgent steering direction of the experiment participant, references were set for the steering wheel and steering rate because the steering wheel turns even without the operation of the driver due to the nature of the steering wheel. Similarly to D'Addario and Donmez (2019), the criteria for determining whether the experiment participant steered was set to a change in the steering wheel of 10

degrees or more and 15 degrees/s or more for the steering rate after hazard onset. Fig. 6(c) shows the steering wheel angle of an experiment participant, and it can be observed that after perception of a hazardous situation, the participant steered to the left(+) to avoid collision. The time consumed to reach the established steering criteria is SRT_p .

Driver Reaction Time from Perception (DRT_p)

The driver reaction time from perception is the reaction time of the experiment participant for either braking or steering, whichever occurs first, after hazard onset perception. This reaction time was defined to measure the fastest reaction time of the driver (Attalla et al., 2018). For example, if the driver steers first and then brakes like the experiment participant data shown in Fig. 6, SRT_p becomes DRT_p .

2.5. Apparatus & data acquisition

Equipment used for the experiment were a full-scale driving simulator and Smart Eye Pro eye tracker as shown in Fig. 7. The simulator was based on the 2016 Hyundai LF Sonata automobile and shakers were installed on all the wheels for the 3DOF motion of roll, heave, and pitch so that vibrations corresponding to vehicle speed, road surface characteristics, and engine operation can be delivered to the passenger. Inside the cabin, 3 cameras were installed for observation of the steering wheel status and whether the experiment participant was looking forward, the driving environment from the perspective of the experiment participant, and whether the pedals were being operated. The driving environment was projected to the driver using 3 projectors of 1920x1200(WUXGA) video resolution, and 5.1 channel speakers were installed within the cabin to effectively deliver sounds from the virtual environment at their corresponding relative directions. The simulator software SCANeR studio (ver. 1.8) of AV Simulation was used to reproduce the 4 hazardous situations, and the data pertaining to the hazard onset time, accelerator pedal displacement, brake pedal effort force, steering wheel angle, and steering wheel angular velocity were synchronized for measurement. The Smart Eye Pro eye tracker of Smart Eye AB is composed of 3 cameras and an infrared sensor to track the angle, direction, and movement of the head and line of sight for 6DOF at a sampling rate of 60 Hz and with an accuracy of 0.5 degrees. The eye tracker was installed above the dashboard within the cabin to investigate SL, a dependent variable, after measuring the direction, position, and movement of the gaze of the experiment participant before and after hazard onset.

2.6. Procedure

Participants were instructed regarding the overall experiment and participants completed experiment consent forms. Afterwards, participants entered the vehicle, the eye tracker was adjusted for calibration, and participants drove for practice for around 10 min to become familiar with the simulator.

All participants were instructed to drive at 60 km/h and 80 km/h, but the order was randomly assigned to prevent a learning effect. The order of hazardous situations also varied by driving speed. The hazardous situation order for 60 km/h was 1) driving over the

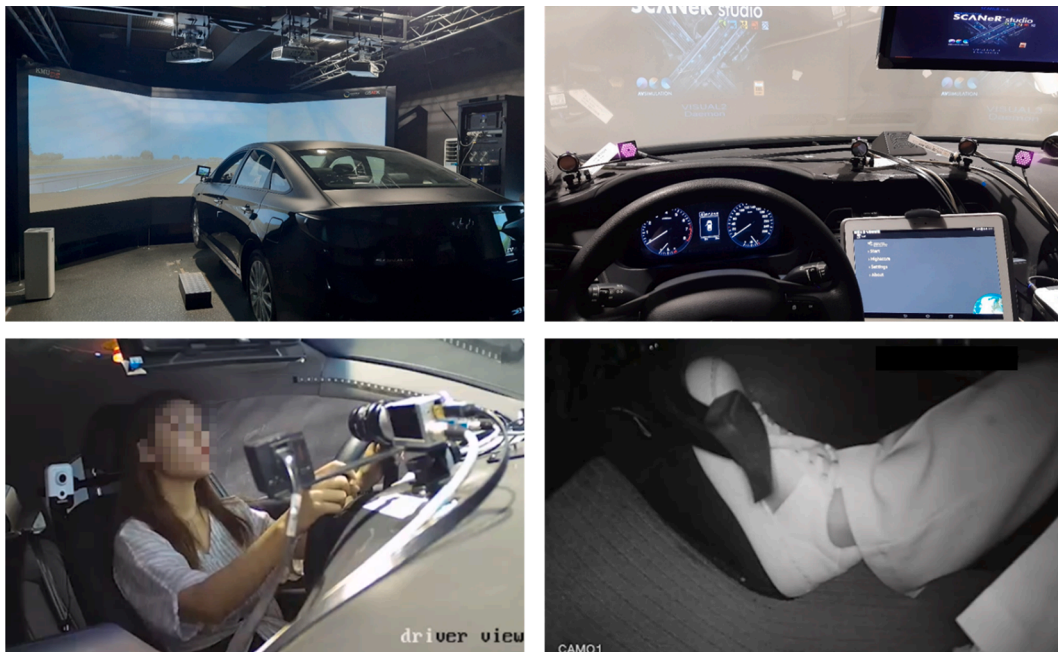


Fig. 7. Simulator exterior (top left), cabin interior with eye tracker installation (top right), an experiment participant in the driver's seat (bottom left), pedal operation by an experiment participant (bottom right).

centerline, 2) pedestrian jaywalking, 3) cutting in, and 4) intersection traffic signal violation, and the situation order for 80 km/h was 1) intersection traffic signal violation, 2) cutting in, 3) driving over the centerline, and 4) pedestrian jaywalking, as shown in Fig. 8. The driving time was around 10 min each at 60 km/h and at 80 km/h.

3. Results & discussion

SCANer studio (ver 1.8) was used to log data about the vehicle speed, accelerator pedal operation, brake pedal operation, steering wheel operation, and gaze of the experiment participant. Then, the dependent variables were obtained using Matlab (R2018a) of MathWorks. SL was set to 0 when the gaze of the experiment participant moved to the region where the hazard was located before hazard onset (D'Addario and Donmez, 2019). In the data analysis, the data with SL equal to 0 and data without DRT_p were excluded because it is difficult to accurately measure the timepoint of hazardous situation perception when SL is 0, and the objective of this study was to investigate the rapid reaction time for accident avoidance after perception of a hazardous situation by the experiment participant. Moreover, data without DRT_p was excluded because it was not possible to determine the reaction time of the experiment participant. The following inferential statistics used Jamovi (ver 1.1.9.0). T-tests were used to analyze reaction time by speed and gender, and one-way ANOVA was applied to analyze reaction time by age group and hazardous situation. The Tukey test was employed for a post-hoc analysis.

3.1. Age group

Table 3 shows the descriptive statistics values according to age group and Fig. 9 shows the average, standard deviation, and significance of the reaction time according to age group. Statistical analyses according to age group revealed that there were no

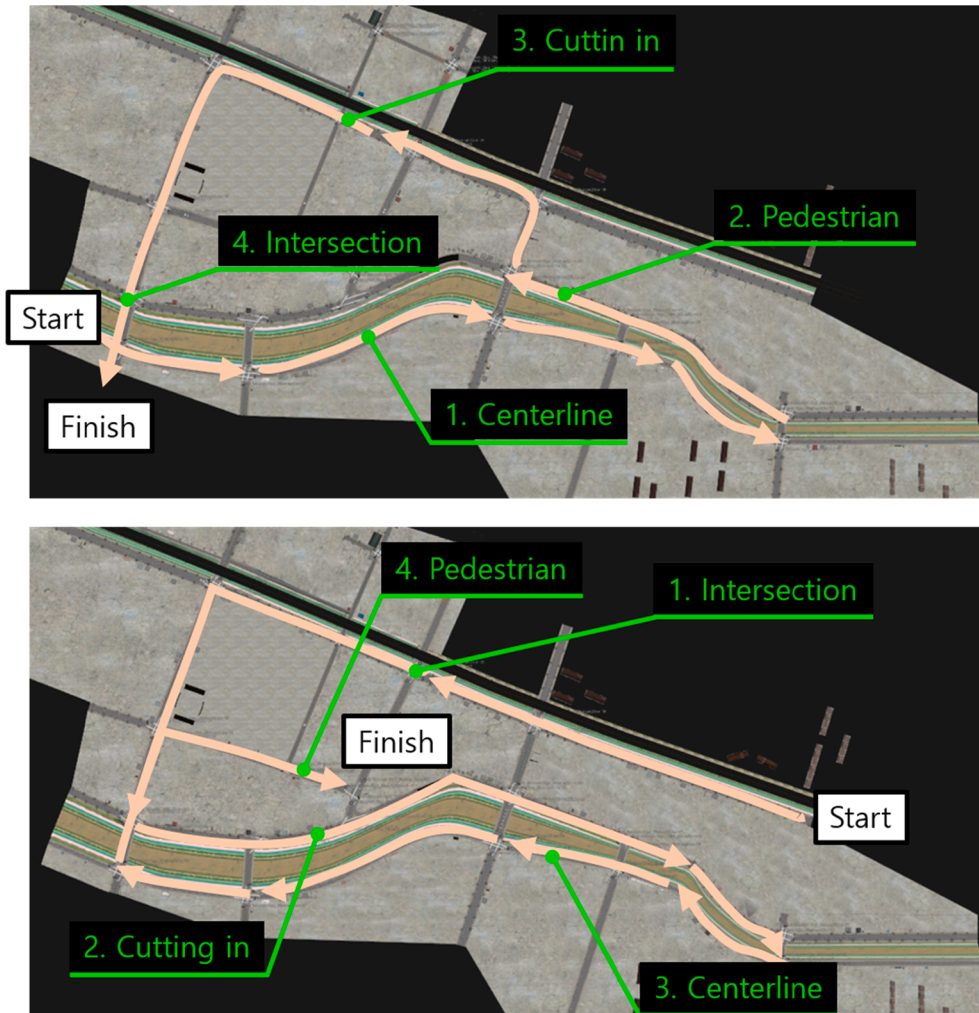


Fig. 8. Route and the order of hazardous situations. (60 km/h (top), 80 km/h (bottom)).

significant differences for SL($F(2, 483) = 1.49, p = .227$), while the average value for the 20 s, 30 s, and 40 s age groups were 0.711 s, 0.710 s, and 0.769 s, with all values falling between 0.7 s and 0.8 s. Significant differences were observed for BRT_p($F(2, 453) = 11.88, p < .001$) and the post-hoc analysis resulted in the 40 s age group having shorter BRT_p compared to that of the 20 s and 30 s age groups. The BRT_p average of the 40 s age group was 0.382 s, which was 0.119 s shorter than the 20 s age group BRT_p average of 0.501 s ($t(453) = 3.94, p < .001$) and 0.138 s shorter than the 30 s age group BRT_p average of 0.520 s ($t(453) = 4.42, p < .001$).

Significant differences were also found for SRT_p($F(2, 286) = 6.02, p = .003$) and the post-hoc analysis showed that the 40 s age group steered faster after hazardous situation perception than the 20 s and 30 s age groups. The 40 s age group average was 0.615 s, which was 0.189 s shorter than the 20 s age group average of 0.804 s ($t(286) = 2.98, p < .009$) and 0.207 s shorter than the 30 s age group average of 0.822 s ($t(286) = 2.90, p < .011$). DRT_p($F(2, 483) = 13.34, p < .001$) also showed significant differences, and the post-hoc analysis resulted in significant differences between the 40 s and 20 s age groups and between the 40 s and 30 s age groups. The DRT_p average for the 40 s age group was 0.355 s, which was 0.105 s shorter than the 20 s age group average of 0.460 s ($t(483) = 3.77, p < .001$) and 0.143 s shorter than the 30 s age group average of 0.498 s ($t(483) = 4.92, p < .001$).

Summarizing the above results, there were no significant differences between the age groups for SL, and all BRT_p, SRT_p, and DRT_p for the 40 s age group were shorter than those of the 20 s and 30 s age groups. These results appear to show a relationship with the driving experience of the driver, where the driving experience span increased with older age groups. The driving experience averages of the 20 s, 30 s, and 40 s age groups were 2.52, 9.93, and 18.4 years. However, the driving experience span difference between the 20 s and 30 s age groups did not have a significant impact on the reaction times, and this was thought to reflect the actual amount of driving experience of the participant, in addition to the number of years of driving experience. Most of the participants in their 40 s were on-call drivers, so due to the nature of their job, they drove on city roads more than ordinary citizens. Therefore, the 40 s age group participants will have experienced more hazardous situations on the road more regularly than the participants in the 20 s and 30 s age groups, resulting in increased response ability and shorter reaction times.

3.2. Gender

The descriptive statistics values according to gender are shown in Table 4, and the average, standard deviation, and significance values of the reaction time according to gender are shown in Fig. 10. Statistical analyses according to gender showed that there were no significant differences for SL ($t(484) = 0.89, p = .375$) where the averages for male and female participants were 0.744 s and 0.715 s, respectively. A significant difference was found for BRT_p ($t(454) = -2.57, p = .011$) where the averages for male and female participants were 0.435 s and 0.500 s, respectively. The male participants applied the brakes 0.065 s faster on average than the female participants. A significant difference was observed for SRT_p ($t(287) = -2.78, p = .006$) where the averages for male and female participants were 0.671 s and 0.829 s, respectively. The male participants steered 0.158 s faster on average than the female participants. A significant difference was also found for DRT_p ($t(484) = -2.67, p = .008$) where the averages for male and female participants were 0.406 s and 0.469 s, respectively, meaning the response time was shorter by 0.063 s for male participants in comparison to the female participants.

Thus, there was no significant difference in the time to perceive a hazardous situation while driving between the male and female participants, but the reaction time using steering or braking after perception was shorter for the male participants than the female participants. In a study on the differences in lower body kinematics and ground reaction forces depending on gender, Spiteri et al. (2014) revealed that the reaction time for males was shorter than for females, and it was determined that such differences can also result in significant differences for BRT_p and SRT_p. In a study of driver-injury severities by gender under wet road-surface conditions, Islam and Mannering (2021) showed that there were differences in perception of hazardous situations and reaction times by gender.

Table 3
Descriptive statistics of perception reaction time by age group.

Age group	DV (s)	SL	BRT _p	SRT _p	DRT _p
20 s	N	172	159	106	172
	Mean	0.711	0.501	0.804	0.460
	SD	0.341	0.283	0.486	0.237
	Min	0.020	0.100	0.100	0.100
	Median	0.645	0.430	0.665	0.420
	Max	1.780	2.220	2.410	1.520
30 s	N	148	140	71	148
	Mean	0.710	0.520	0.822	0.498
	SD	0.370	0.301	0.482	0.295
	Min	0.020	0.010	0.100	0.040
	Median	0.645	0.465	0.740	0.440
	Max	1.750	1.430	1.940	1.430
40 s	N	166	157	112	166
	Mean	0.769	0.382	0.615	0.355
	SD	0.354	0.213	0.447	0.240
	Min	0.010	0.070	0.020	0.020
	Median	0.725	0.340	0.500	0.300
	Max	1.840	1.150	1.930	1.850

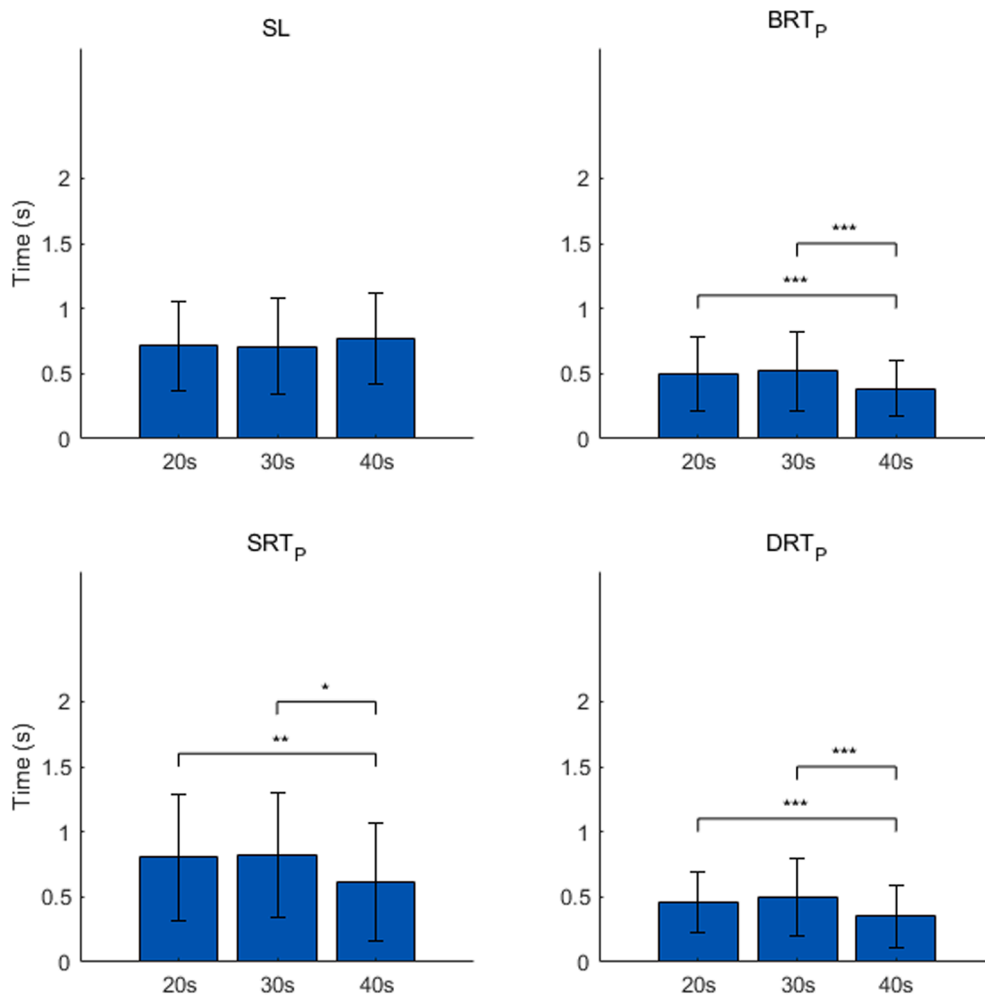


Fig. 9. Perception reaction time according to age group with the error bars showing standard deviations (* $p < .05$, ** $p < .01$, *** $p < .001$).

Table 4

Descriptive statistics of perception reaction time by gender.

Gender	DV (s)	SL	BRT _p	SRT _p	DRT _p
Male	N	258	239	172	258
	Mean	0.744	0.435	0.671	0.406
	SD	0.339	0.261	0.428	0.242
	Min	0.010	0.040	0.100	0.040
	Median	0.705	0.370	0.560	0.360
	Max	1.750	2.220	1.940	1.380
Female	N	228	217	117	228
	Mean	0.715	0.500	0.829	0.469
	SD	0.373	0.283	0.532	0.282
	Min	0.010	0.070	0.020	0.020
	Median	0.635	0.430	0.730	0.410
	Max	1.840	1.520	2.410	1.850

3.3. Speed

Table 5 shows the descriptive statistics values according to speed, and Fig. 11 shows the average, standard deviation, and significance values of the reaction time according to speed. The statistical analyses according to speed showed no significant difference for SL ($t(484) = -0.96$, $p = .340$) where the average values for 60 km/h and 80 km/h were 0.716 s and 0.747 s, respectively. There was no significant difference for BRT_p ($t(454) = 0.79$, $p = .432$) as well, and the average values for 60 km/h and 80 km/h were 0.476 s and 0.456 s, respectively. A significant difference was observed for SRT_p ($t(287) = -2.78$, $p = .006$) where the average values for 60 km/h

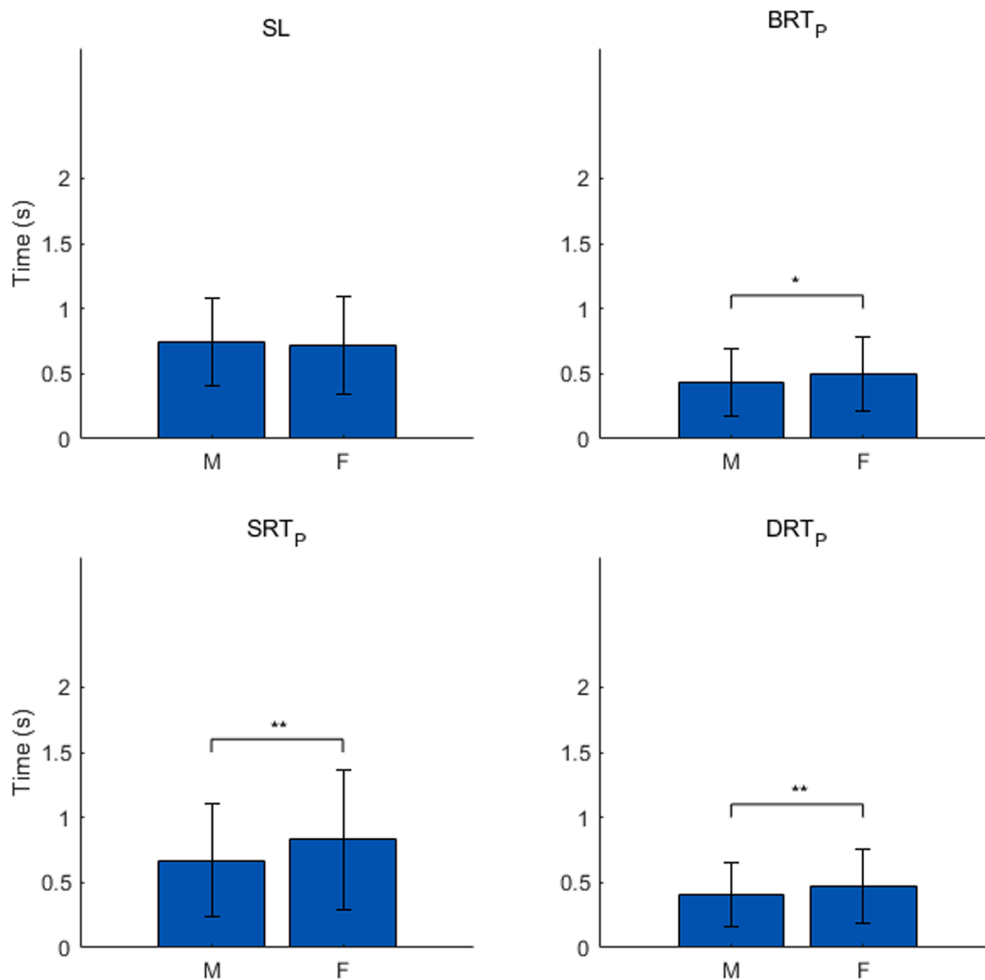


Fig. 10. Perception reaction time according to gender with the error bars showing standard deviations (* $p < .05$, ** $p < .01$, *** $p < .001$).

Table 5

Descriptive statistics of perception reaction time by speed.

Speed	DV (s)	SL	BRT _p	SRT _p	DRT _p
60 km/h	N	251	230	129	251
	Mean	0.716	0.476	0.650	0.443
	SD	0.378	0.301	0.437	0.285
	Min	0.020	0.040	0.020	0.020
	Median	0.670	0.410	0.570	0.390
	Max	1.840	2.220	1.940	1.850
80 km/h	N	235	226	160	235
	Mean	0.747	0.456	0.804	0.428
	SD	0.328	0.242	0.501	0.238
	Min	0.010	0.070	0.110	0.070
	Median	0.670	0.405	0.675	0.380
	Max	1.780	1.450	2.410	1.380

and 80 km/h were 0.650 s and 0.804 s, respectively, so the average value for the speed of 60 km/h was shorter by 0.154 s. There was no significant difference for DRT_p ($t(484) = 0.62$, $p = .535$) where the averages for 60 km/h and 80 km/h were 0.443 s and 0.428 s, respectively. Among the perception reaction times, only SRT_p showed a significant difference, and according to Rogé et al. (2004), the useful visual field decreases as speed increases. Given the nature of steering avoidance, the experiment participant has to visually find a place to avoid, and when driving at the speed of 80 km/h, the useful visual field is smaller than when driving at 60 km/h, so the time consumed in determining whether to carry out steering avoidance will be longer. Moreover, there were no significant differences for perception reaction times other than SRT_p, showing that the speeds of 60 km/h and 80 km/h are not factors that significantly impact

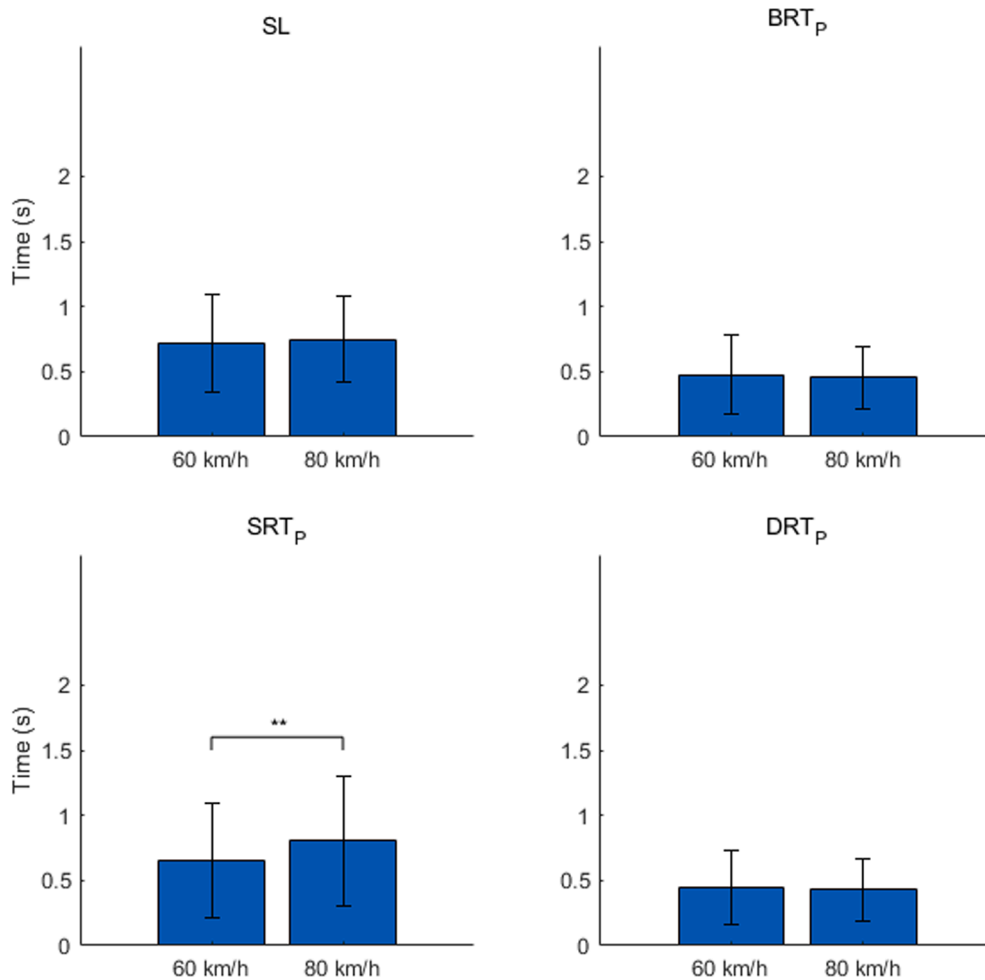


Fig. 11. Perception reaction time according to speed with the error bars showing standard deviations (* $p < .05$, ** $p < .01$, *** $p < .001$).

the perception reaction time.

3.4. Hazardous situations

Analysis of the reaction time according to the hazardous situation resulted in significant differences for all SL ($F(3, 482) = 117.25$, $p < .001$), BRT_p ($F(3, 452) = 6.48$, $p < .001$), SRT_p ($F(3, 285) = 15.87$, $p < .001$), and DRT_p ($F(3, 482) = 5.24$, $p = .001$). The average, standard deviation, and significance values according to the hazardous situation are shown in Fig. 12 and Table 6 presents descriptive statistics values of the reaction time according to the hazardous situation.

The SL average values for driving over the centerline, pedestrian jaywalking, cutting in, and intersection traffic signal violation were 0.762 s, 0.549 s, 1.090 s, and 0.532 s. Post-hoc analysis resulted in the driving over the centerline being 0.213 s longer on average than the pedestrian jaywalking ($t(482) = 5.75$, $p < .001$), 0.328 s shorter on average than cutting in ($t(482) = -9.13$, $p < .001$), and 0.230 s longer on average than the intersection traffic signal violation ($t(482) = 6.63$, $p < .001$). The pedestrian jaywalking was 0.541 s shorter on average than the cutting in ($t(482) = 15.31$, $p < .001$).

The significant difference results showed that the order of SL length was the pedestrian jaywalking and intersection traffic signal violation, driving over the centerline, cutting in, and this result appears to be due to the influence of the varying hazard onset locations. For example, the pedestrian jaywalking and intersection traffic signal violation situations have the pedestrian and signal violating vehicle appear in the direction perpendicular to the experiment participant, so the participant can perceive the hazardous situation relatively quickly. On the other hand, the driving over the centerline and cutting in vehicles appear gradually in directions diagonal to the forward direction of the experiment participant, so more time can be consumed in the perception of the hazardous situation.

Post-hoc analysis of BRT_p showed that the average value of 0.397 s for the intersection traffic signal violation was 0.091 s shorter than the average value of 0.488 s for the pedestrian jaywalking situation ($t(452) = -2.65$, $p = .041$) and 0.148 s shorter than the average value of 0.545 s for the cutting in situation ($t(452) = -4.30$, $p < .001$). The average value was 0.461 s for the driving over the centerline situation, which showed no significant differences with the other hazardous situations.

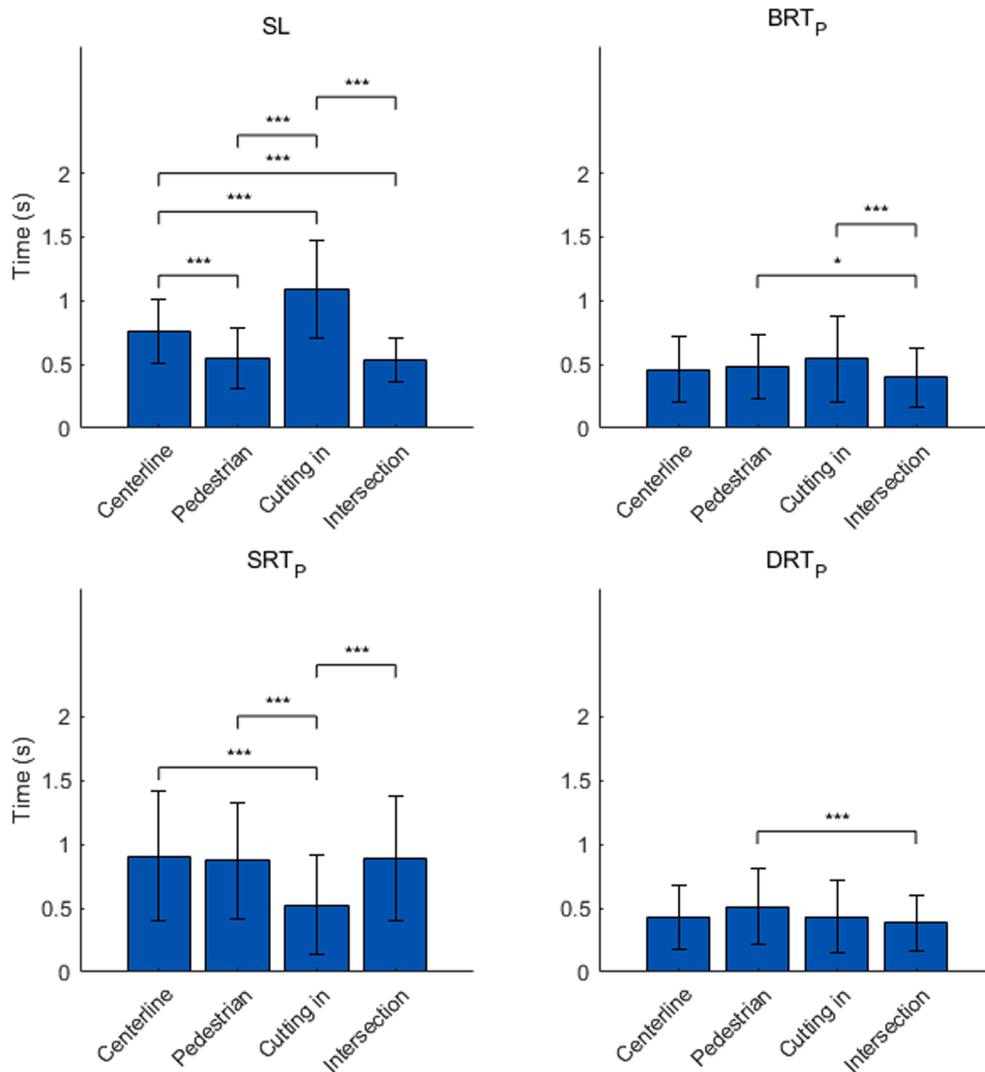


Fig. 12. Perception reaction time according to the hazardous situation with the error bars showing the standard deviations (* $p < .05$, ** $p < .01$, *** $p < .001$).

The post-hoc analysis result of SRT_p showed that the results for cutting were significantly different than the other hazardous situations. The average value of 0.524 s for the cutting in situation was 0.380 s shorter than the average value of 0.904 s for the driving over the centerline ($t(285) = 5.82$, $p < .001$), 0.346 s shorter than the average value of 0.870 s for the pedestrian jaywalking ($t(285) = -4.661$, $p < .001$), and 0.362 s shorter than the average value of 0.886 s for the intersection traffic signal violation ($t(285) = 4.50$, $p < .001$). In the earlier discussed SL, the cutting in situation value was longer than the other hazardous situations. The participants perceived the hazardous situation that much later, so there was little time for avoidance. To avoid an accident with the cutting in vehicle that approaches or enters the lane of the subject vehicle, steering is needed more than braking, thus, it was thought that the SRT_p for this hazardous situation was shorter than the other situations.

For DRT_p , the average value of 0.513 s for the pedestrian jaywalking situation was 0.130 s longer than the average value of 0.383 s for the intersection traffic signal violation ($t(482) = -3.95$, $p < .001$). The average values for the driving over the centerline and cutting in situations were 0.532 s and 0.432 s, respectively, and these 2 hazardous situations showed no statistically significant differences with the other hazardous situations. The differences between the DRT_p averages and BRT_p averages for the driving over the centerline, pedestrian jaywalking, and intersection traffic signal violation situations were within 0.025 s and their differences with SRT_p were between 0.357 and 0.503 s, showing that the averages of the DRT_p and BRT_p were similar, and for these hazardous situations, the participants avoided the hazards by braking first rather than steering.

3.4.1. Frequency analysis of avoidance strategy by hazardous situations

After hazard onset and perception by the driver, avoidance can be attempted by steering and braking. In this study, to analyze the

Table 6

Descriptive statistics of reaction time according to the hazardous situation.

Hazardous situations	DV (s)	SL	BRT _p	SRT _p	DRT _p
Driving over the centerline	N	104	104	75	104
	Mean	0.762	0.461	0.904	0.431
	SD	0.254	0.260	0.506	0.252
	Min	0.010	0.040	0.100	0.040
	Median	0.835	0.385	0.930	0.376
Pedestrian jaywalking	Max	1.300	1.330	2.410	1.290
	N	110	104	51	110
	Mean	0.549	0.488	0.870	0.513
	SD	0.237	0.252	0.450	0.293
	Min	0.010	0.100	0.140	0.100
Cutting in	Median	0.580	0.430	0.740	0.440
	Max	1.170	1.520	1.860	1.850
	N	127	103	122	127
	Mean	1.090	0.545	0.524	0.432
	SD	0.380	0.335	0.385	0.281
Intersection traffic signal violation	Min	0.020	0.150	0.100	0.100
	Median	1.130	0.520	0.425	0.360
	Max	1.840	2.220	1.940	1.430
	N	145	145	41	145
	Mean	0.532	0.397	0.886	0.383
	SD	0.173	0.231	0.488	0.216
	Min	0.030	0.070	0.020	0.020
	Median	0.540	0.360	0.830	0.360
	Max	1.130	1.320	2.070	1.320

avoidance method according to the hazardous situation, the method of avoidance was divided into steering, braking, braking after steering, steering after braking, and steering and braking simultaneously. Chi-squared analysis revealed that there were significant differences in the avoidance method depending on the hazardous situation ($\chi^2 = 247$, $p < .001$). Table 7 shows the avoidance method according to the hazardous situation. For the driving over the centerline situation, 87.5 % of the participants put on the brakes first for avoidance, with 29 cases of braking (27.9 %) and 62 cases of braking and then steering (59.6 %). For the pedestrian jaywalking situation, 90 % of the participants avoided by braking first with 59 cases of braking (53.6 %) and 40 cases of steering after braking (36.4 %). In the cutting in situation, 64.6 % of the participants steered first for avoidance with 24 cases of steering (18.9 %) and 58 cases of braking after steering (45.7 %). In the intersection traffic signal violation, 94.5 % of the participants avoided by braking first, with 104 cases of braking (71.7 %) and 33 cases of steering after braking (22.8 %). Therefore, except for the cutting in hazardous situation, most participants were found to avoid the hazard by braking first upon hazard onset.

3.4.2. 85th percentile of reaction time

Literature related to the conventional perception reaction time has frequently reported the average as well as various percentiles, especially the 85th percentile (Olson and Sivak, 1986; Olson, 1989; Wong and Goh, 2000). This is to consider the majority of drivers (Olson, 1989). Since the application of the wrong reaction time can lead to misleading conclusions regarding potential accident avoidance, an appropriate value has to be applied, and in this study, the 85th percentile that was most frequently used in previous research and literature was used. As shown in 3.4.1, except for the cutting in situation, most participants avoided the hazards by first braking, so the 85th percentile of BRT_p was investigated along with the 85th percentile of DRT_p to investigate the preferential reaction time of the participants.

Fig. 13 shows the BRT_p scatter plot and histogram for each hazardous situation, and the points corresponding to the 85th percentile are marked. The 85th percentile of each hazardous situation was 0.659 s for the driving over the centerline, 0.737 s for the pedestrian jaywalking, 0.700 s for the cutting in, and 0.550 s for the intersection traffic signal violation. So, the 85th percentile was in the range of 0.550 – 0.737 s. Also, combining the 4 hazardous situations, the 85th percentile was 0.660 s. Fig. 14 shows the DRT_p scatter plot and histogram for each hazardous situation, and the points corresponding to the 85th percentile are marked. The DRT_p 85th percentile was 0.650 s for the driving over the centerline, 0.800 s for the pedestrian jaywalking, 0.660 s for the cutting in, and 0.550 s for the

Table 7

Avoidance method frequency according to the hazardous situation with percentage values in parentheses.

Hazardous situations	Avoidance strategy					Total
	Steer	Brake	Steer after brake	Brake after steer	Steer and brake ^a	
Driving over the centerline	0 (0)	29 (27.9)	12 (11.5)	62 (59.6)	1 (1.0)	104
Pedestrian jaywalking	6 (5.5)	59 (53.6)	4 (3.6)	40 (36.4)	1 (0.9)	145
Cutting in	24 (18.9)	5 (3.9)	58 (45.7)	40 (31.5)	0 (0.0)	127
Intersection traffic signal violation	0 (0.0)	104 (71.7)	8 (5.5)	33 (22.8)	0 (0.0)	110

^a Steering and braking simultaneously.

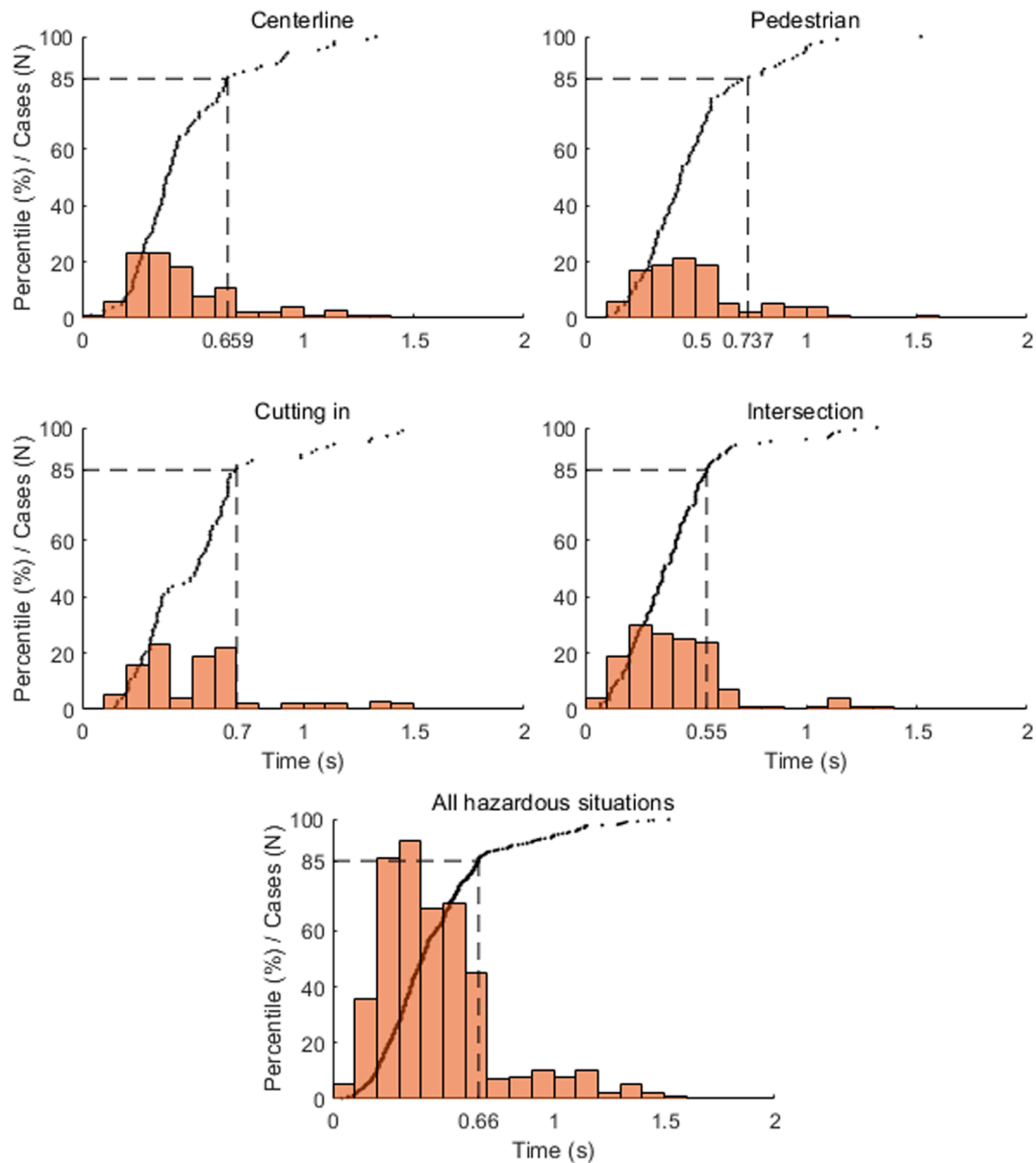


Fig. 13. Cumulative distribution and 85th percentile of BRT_p by hazardous situations.

intersection traffic signal violation. So, the DRT_p was in the range of 0.550 – 0.800 s. Combining all the hazardous situations, the 85th percentile of DRT_p was 0.646 s. This means that most of the drivers perceived, determined, and steered or braked within 0.646 s after perception of the hazardous situation. Since most participants braked first to avoid collision, the 85th percentiles of BRT_p and DRT_p did not differ significantly. As can be observed in Fig. 13 and Fig. 14, BRT_p and DRT_p increased consistently for all the hazardous situations and then rapidly increased at the 85th percentile.

4. Conclusion

The objective of this study was to determine the reaction time applicable for traffic accident analysis. A full-scale driving simulator was used to reproduce 4 hazardous situations and carry out the experiments. A total of 93 male and female subjects with ages ranging from their 20 s to 40 s participated in the experiment, where they drove at speeds of 60 km/h and 80 km/h. The reaction times according to the speed and hazardous situation were obtained, significant differences were investigated, and the 85th percentile, which is the time that most participants can react in, was examined.

While significant differences in saccade latency (SL) were observed according to age group, gender, and speed, the brake reaction

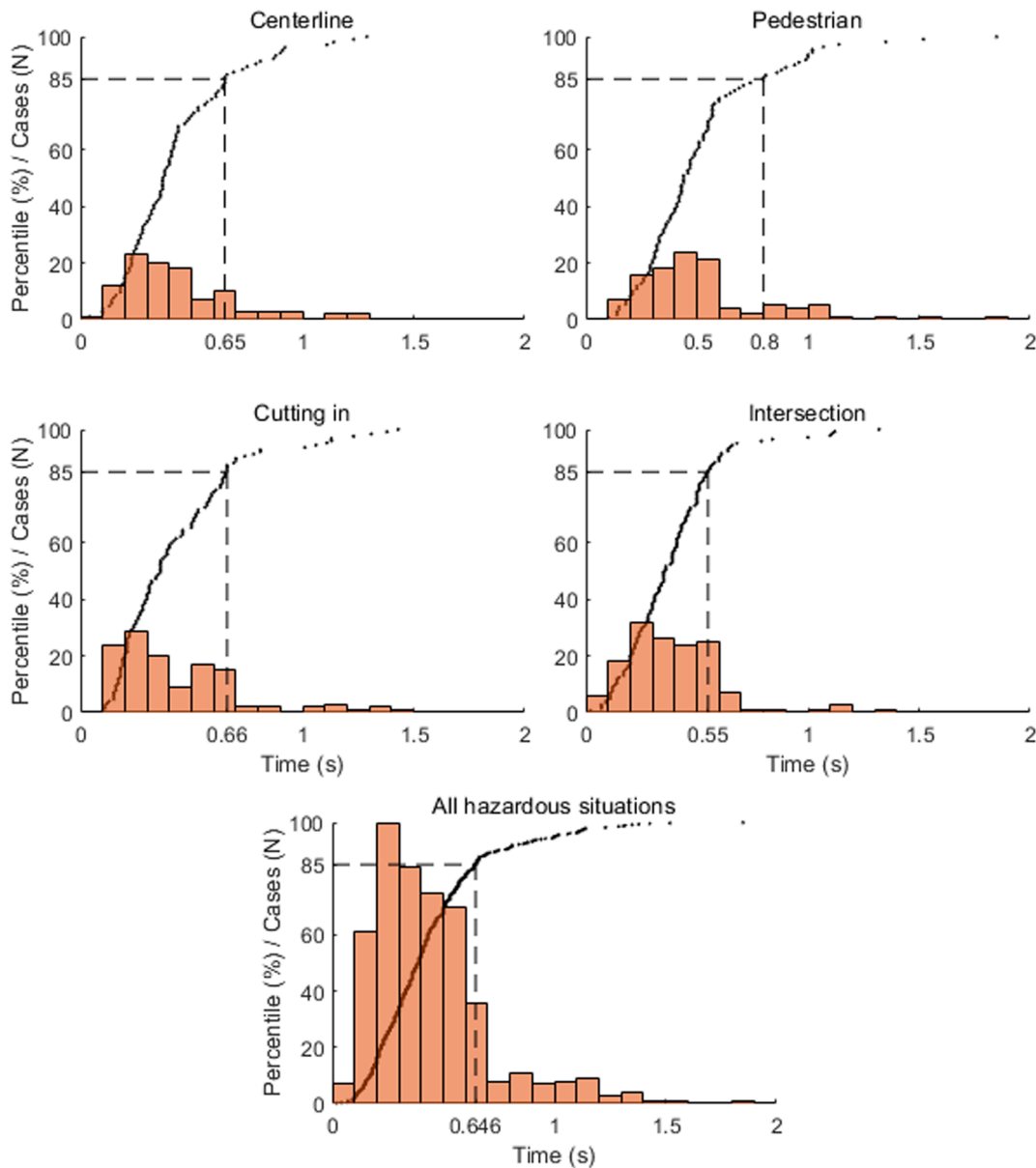


Fig. 14. Cumulative distribution and 85th percentile of DRT_p by hazardous situations.

time from perception (BRT_p), steer reaction time from perception (SRT_p), and driver reaction time from perception (DRT_p) were shorter for the 40 s age group compared to that of the 20 s and 30 s age groups and shorter for the male participants than the female participants. Most participants were found to brake first to avoid an accident in the hazardous situations, except for the cutting in situation. The 85th percentile of BRT_p was 0.659 s for the driving over the centerline situation, 0.737 s for the pedestrian jaywalking situation, 0.700 s for the cutting in situation, and 0.550 s for the intersection traffic signal violation situation, so the range was 0.550 – 0.737 s. Looking at all the situations together, the 85th percentile was 0.660 s. The 85th percentile of DRT_p was 0.650 s for the driving over the centerline, 0.800 s for the pedestrian jaywalking, 0.660 s for the cutting in, and 0.550 s for the intersection traffic signal violation, so the range was 0.550 – 0.800 s. Combining all the hazardous situations gave the result of 0.646 s. As such, when analyzing accidents similar to the hazardous situations reproduced in this study, the above reaction times of drivers may be considered.

When analyzing reaction times of drivers in actual traffic accidents, it is necessary to examine the time taken for drivers to become aware of such situations and to avoid accidents by steering or braking. The values obtained in past research are difficult to apply to actual scenarios as reaction times were measured after the occurrence of hazardous situations, regardless of drivers' perception. This study used an eye tracker to measure reaction time after perception, thereby producing values more useful for accident analysis compared to previous research.

As this study was conducted with participants in their 20 s, 30 s, and 40 s, there is a need to acquire additional data regarding elderly drivers by recruiting participants in their 50 s and beyond. Since this study was limited to daytime situations, further analysis may be performed by designing nighttime situations, and comparing the results by time of day. The values obtained from reproduction of hazardous situations in the simulator should also be compared against real-world values to assess their reliability. Such research is expected to provide valuable insight into the analysis of potential accident avoidance for drivers in the near future.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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